



BSc COMPUTER SCIENCE

CM1040 Web Development

End of term coursework

For this project, you are to create a website which is about the history of the internet in a country of your choosing, during your own lifetime. It is up to you how you interpret this, but some examples of areas of interest are: early ISPs and commercial internet provision and/or academic networks, public WIFI, and so on. Your website should have at least three pages, and some of the content on the pages should be rendered from JSON data files using the template engine encountered in the module.

The coursework is worth 70% of your final mark.

The project should be carried out in several stages:

1) Background research about the internet in your chosen country and project planning [20 marks]

Output of this stage: a plan for the topic to be covered on each page of the website, with a fact sheet and notes. Also, images and other additional media you might want to include in your website. Ensure you are able to cite sources for any media you include such as images and text that you did not create. Also, a plan for the later stages of development which references some of the project planning techniques covered in the course.

Expectation: you are expected to thoroughly research the history of the internet in your chosen country, focusing on key events and developments during your lifetime. Use this research to create a clear and structured plan for your website, detailing the topic to be covered on each page. Gather relevant media (e.g., images, videos) and ensure all sources are properly cited. Your project plan should show how you will approach the development stages, applying project management techniques covered in the course.

2) Website layout prototype development, including responsive layouts for different devices. Show designs to other people and show feedback and improvements made [20 marks]

Output of this stage: wireframe prototypes for all pages on the website, with versions of the layouts for desktop and mobile devices. Evidence that you presented the designs

to other people (either other students or anyone else you are in contact with). Evidence that you updated the designs based on the feedback from the other people.

Expectation: you should create detailed wireframes for every page of your website, showing how the content will be structured. Ensure your designs include separate layouts for both desktop and mobile devices, demonstrating responsiveness. Focus on usability and aesthetics, ensuring that the designs meet accessibility and usability standards discussed in the course.

3) Build the pages of the website, using a template engine to convert JSON data into HTML when appropriate. Include JSON data validation code [40 marks]

Output of this stage: HTML, JavaScript and CSS code files for the website.

Expectation: develop a functional and polished website based on your wireframes. Use HTML, CSS, and JavaScript to build the pages, and ensure that some of the content is rendered dynamically using a template engine and JSON data files. Aim for clean, well-organised code that adheres to best practices in web development. You should use the Visual Studio Code functionality seen in the course to serve up the files for your website so the JavaScript code can access the JSON data via HTTP.

Extensions: We will reserve some extra marks for submissions that attempt to go beyond the expectations described above. Here are some suggestions as to how you might attempt to do that. These steps are not essential – you can achieve a high mark without these extensions.

- Access your JSON data via an API: create a REST API server as shown in the book data example project to serve up the data for your website instead of just directly using JSON files. You might go even further and think of a way to have visitors to your site contribute to its content. You might also consider accessing data from a public API if you can find one that is relevant.
- Implement some sort of JSON data validator that checks the JSON data is valid for your purposes before passing it to the template. You should implement some code in your JavaScript layer that checks the needed things are in the JSON data before attempting to render the template.

4) Carry out HTML validation and accessibility tests [10 marks]

Output of this stage: automated reports from the validation and accessibility tools seen in the course. Include reports before and after you fixed the detected problems in your report.

Expectation: run validation and accessibility tests on your website using tools demonstrated in the course. Fix any issues identified in the initial tests and generate updated reports. Provide evidence of your fixes and reflect on the importance of these standards for creating high-quality, inclusive web applications.

5) Write up report and submit website code [10 marks]

Output of this stage: the complete project report and the complete code for the website.

Expectation: submit a comprehensive report that documents your entire project. This should include the expected outputs of the stages above such as your research, planning, development process, testing, and reflections on what you learned. The report should be in **PDF format**. You should submit this file to the appropriate prompt on the Coursera platform.

The report should demonstrate your understanding of the course concepts and explain the decisions you made during development, highlighting how you met the project requirements.

The maximum word count for your report is **1,500 words**. This does not include the source code – you can make that as long as you like. We recommend that you structure your report with the following sections:

- A. **Background research**
- B. **Planning**
- C. **Development process: prototype designs**
- D. **Development process: developing the code**
- E. **Testing: code validation reports and actions taken**
- F. **Reflections on what you learned**

Submit all the website code with clear comments and organisation. The code should be combined into a single **ZIP format file**. **Use standard zip, not 7zip, rar, etc.** You should submit this to the appropriate prompt on the Coursera platform.

[END OF COURSEWORK ASSIGNMENT]